

Ian Leung

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EDUCATION

University of Waterloo

Expected 2029

Bachelor of Computer Science and Finance - Double Major

GPA: 3.8/4.0

- President's Scholarship of Distinction (\$5000), Undergraduate Achievement Award (\$1500)
- Statistics Minor

Relevant Coursework: Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Data Analytics

SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, VictoriaMetrics, PostgreSQL, MongoDB, React, Actions

Tools Linux, Git, Docker, AWS EC2, OpenRouter, Jupyter, Modal, Supabase, Vercel, Unity, Android Studio

WORK EXPERIENCE

Roblox

Jun 2026 – Present

Software Engineer Intern (Network Reliability) | Go, Python

San Mateo, CA

- Built an automated CI/CD pipeline that deploys configuration changes across **5,000+** network devices in global datacenters, cutting deployment time by **96%**
- Designed and orchestrated Temporal workflows for safe, gated deployment of configuration changes, validating changes on canary devices before progressive fleet-wide rollout with **30+** parallel workers
- Identified and fixed a resource leak in an internal network service, preventing API lockout and recovering **7%** CPU utilization across devices

Waterloo Quant Club

Jan 2026 – Present

Software Developer | Rust, Python, TypeScript

Waterloo, ON

- Deployed www.portal.waterlooquantclub.com to AWS EC2 as a Docker Compose stack (Postgres, Flask, React) behind Caddy for automatic TLS, serving **500+** members
- Built a CI/CD pipeline that SSH-deploys on merge to main, rebuilds containers, and gates the release on a post-deploy health check
- Developed an email verification service using AWS SES, sending **100+** emails per day to new members and automating account verification

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework which generates synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Made a logging API to validate test results against baselines, catching **7** software defects before production release

PROJECTS

Data Science Club Website | TypeScript, React, Supabase, HTML, CSS

- Maintained www.uwdatascience.ca, UWaterloo's data science club website serving **600+** members
- Replaced the executive team's manual term-start **Google Form** with an in-app onboarding flow backed by a normalized **Supabase/Postgres** schema that autofills profile and role data from existing records

Pharmagotchi - Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built a native **Android** health-companion app where a virtual pet responds to missed medication doses, concerning vitals, and skipped logging, backed by an offline **Room** database and **WorkManager** background reminders
- Integrated **LLMs via OpenRouter** to power a context-aware chat assistant and auto-build each user's tracked metrics, parsing their conditions and medications into structured **JSON** vital signs

PeerAssist - Full-Stack Peer Reviewing | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for 20+ students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin